HarnessAgent API Reference

Production-grade multi-agent orchestration harness

GET /health

Health

Aggregate health check for all service dependencies.

Returns: 200 with status="ok" if all services healthy. 200 with status="degraded" if any service is unhealthy (but process is alive).

Responses

Code	Description
200	Successful Response

GET /health/ready

Readiness

Kubernetes readiness probe.

The service is ready when Redis (our primary state store) is reachable.

Returns: 200 if ready, 503 if not ready.

Responses

Code	Description
200	Successful Response

GET /health/live

Liveness

Kubernetes liveness probe.

Always returns 200 as long as the process is alive.

Returns: 200 always.

Responses

Code	Description
200	Successful Response

GET /metrics

Metrics

Prometheus metrics endpoint in text exposition format.

Returns: text/plain with Prometheus metrics.

Responses

Code	Description
200	Successful Response

POST /runs

Create Run

Create a new agent run and enqueue it for a worker.

Args: body: Run creation payload. tenant_id: Extracted from JWT. redis: Redis client from app state.

Returns: 201 with the created RunRecord.

Raises: 400 if agent_type is not recognised.

Parameters

Name	In	Required	Type	Description
`X-API-Key`	header	No	string	

Request Body

Requires `CreateRunRequest` JSON object.

Responses

Code	Description
201	Successful Response
422	Validation Error

GET /runs

List Runs

List runs for the authenticated tenant.

Args: limit: Maximum number of runs to return (1-100). offset: Pagination offset. tenant_id: Extracted from JWT. redis: Redis client.

Returns: 200 with list of RunRecords, newest first.

Parameters

Name	In	Required	Type	Description

`limit`	query	No	integer	
`offset`	query	No	integer	
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

GET /runs/{run_id}

Get Run

Retrieve a run by ID.

Args: run_id: The run identifier. tenant_id: Extracted from JWT. redis: Redis client.

Returns: 200 with RunRecord.

Raises: 404 if run not found. 403 if run belongs to a different tenant.

Parameters

Name	In	Required	Type	Description
`run_id`	path	Yes	string	
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

DELETE /runs/{run_id}

Cancel Run

Cancel a run that is pending or running.

Args: run_id: The run to cancel. tenant_id: Extracted from JWT. redis: Redis client.

Returns: 204 on success.

Raises: 404 if run not found. 403 if run belongs to another tenant.

Parameters

Name	In	Required	Type	Description
`run_id`	path	Yes	string	
`X-API-Key`	header	No	string	

Responses

Code	Description
204	Successful Response
422	Validation Error

GET /runs/{run_id}/stream

Stream Run Events

Stream run events as Server-Sent Events (SSE).

Subscribes to the Redis event stream for *run_id* and forwards events as SSE until the run completes or the timeout expires.

Each event is a JSON-encoded StepEvent payload::

```
data: {"event_type": "token_delta", "payload": {"delta": "Hello"}, ...}
```

Args: run_id: The run to stream events from. timeout: Maximum seconds to keep the connection open (default 300).

Returns: text/event-stream response.

Raises: 404 if run not found. 403 if run belongs to another tenant.

Parameters

Name	In	Required	Type	Description
`run_id`	path	Yes	string	
`timeout`	query	No	number	
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

GET /runs/{run_id}/steps

Stream Run Steps

Stream real-time step events for a run via Server-Sent Events.

The stream uses a Redis pub/sub channel keyed by run_id. Workers publish StepEvent JSON objects to this channel as the agent executes. A `[DONE]` sentinel is sent when the run completes or the connection times out.

Args: run_id: The run to stream. request: FastAPI request object (for disconnect detection). tenant_id: Extracted from JWT. redis: Redis client.

Returns: EventSourceResponse with text/event-stream content type.

Raises: 404 if the run is not found. 503 if the SSE library is not available.

Parameters

Name	In	Required	Type	Description
`run_id`	path	Yes	string	
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

POST /memory/remember

Remember

Store a text entry in long-term memory.

Args: body: The text and metadata to store. tenant_id: From JWT. memory: MemoryManager.

Returns: 201 with the ID of the stored entry.

Parameters

Name	In	Required	Type	Description
`X-API-Key`	header	No	string	

Request Body

Requires `RememberRequest` JSON object.

Responses

Code	Description
201	Successful Response

422	Validation Error

POST /memory/recall

Recall

Retrieve relevant memories via semantic search.

Args: body: Query, k, and optional filter. tenant_id: From JWT. memory: MemoryManager.

Returns: List of MemoryEntry objects ordered by relevance.

Parameters

Name	In	Required	Type	Description
`X-API-Key`	header	No	string	

Request Body

Requires `RecallRequest` JSON object.

Responses

Code	Description
200	Successful Response
422	Validation Error

GET /memory/graph

Query Graph

Retrieve knowledge graph context for a query.

Performs entity extraction on the query, traverses the graph, and returns the discovered paths and rendered context string.

Args: query: Natural language query. max_hops: Maximum graph traversal depth. tenant_id: From JWT. memory: MemoryManager.

Returns: Graph paths and rendered context.

Parameters

Name	In	Required	Type	Description
`query`	query	Yes	string	Search query for graph traversal
`max_hops`	query	No	integer	Maximum graph hops
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

DELETE /memory/session/{run_id}

Clear Session

Clear short-term (session) memory for a specific run.

Args: run_id: The run whose session memory should be cleared. tenant_id: From JWT. memory: MemoryManager.

Returns: 204 on success.

Parameters

Name	In	Required	Type	Description
`run_id`	path	Yes	string	
`X-API-Key`	header	No	string	

Responses

Code	Description
204	Successful Response
422	Validation Error

GET /evals/suites

List Eval Suites

List built-in eval suites available from the operator console.

Responses

Code	Description
200	Successful Response

POST /evals/smoke/run

Run Smoke Eval

Run the built-in single-agent smoke evaluation.

Request Body

Responses

Code	Description
200	Successful Response
422	Validation Error

POST /evals/multi/run

Run Multi Eval

Run the built-in multi-agent handoff evaluation.

Request Body

Responses

Code	Description
200	Successful Response
422	Validation Error

POST /evals/compare

Compare Prompt Eval

Compare baseline and patched prompt labels over the smoke suite.

Request Body

Responses

Code	Description
200	Successful Response
422	Validation Error

GET /improvement/patches

List Patches

List improvement patches, optionally filtered by agent_type and status.

Parameters

Name	In	Required	Type	Description
`agent_type`	query	No	string	
`status`	query	No	string	

`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

POST /improvement/patches/{patch_id}/approve

Approve Patch

Approve a patch and apply it to the active prompt.

Parameters

Name	In	Required	Type	Description
`patch_id`	path	Yes	string	
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

POST /improvement/patches/{patch_id}/reject

Reject Patch

Reject a pending patch.

Parameters

Name	In	Required	Type	Description
`patch_id`	path	Yes	string	
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

POST /improvement/cycle

Trigger Improvement Cycle

Manually trigger a Hermes improvement cycle for an agent type.

Parameters

Name	In	Required	Type	Description
`agent_type`	query	No	string	
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

GET /improvement/errors

List Errors

List recent error records.

Parameters

Name	In	Required	Type	Description
`agent_type`	query	No	string	
`limit`	query	No	integer	
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

GET /improvement/failures/heatmap

Failure Heatmap

Return a heatmap of failure_class counts per agent_type.

Parameters

Name	In	Required	Type	Description
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

GET /hitl/pending

List Hitl Pending

List all pending HITL approval requests for the tenant.

Parameters

Name	In	Required	Type	Description
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

POST /hitl/{request_id}/approve

Approve Hitl

Approve a HITL request.

Parameters

Name	In	Required	Type	Description
`request_id`	path	Yes	string	
`X-API-Key`	header	No	string	

Request Body

Requires `ResolveRequest` JSON object.

Responses

Code	Description

200	Successful Response
422	Validation Error

POST /hitl/{request_id}/reject

Reject Hitl

Reject a HITL request.

Parameters

Name	In	Required	Type	Description
`request_id`	path	Yes	string	
`X-API-Key`	header	No	string	

Request Body

Requires `ResolveRequest` JSON object.

Responses

Code	Description
200	Successful Response
422	Validation Error

GET /prompts/{agent_type}/versions

List Prompt Versions

List all prompt versions for an agent type.

Parameters

Name	In	Required	Type	Description
`agent_type`	path	Yes	string	
`limit`	query	No	integer	
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

POST /prompts/{agent_type}/promote/{version_id}

Promote Prompt

Promote a specific prompt version to active.

Parameters

Name	In	Required	Type	Description
`agent_type`	path	Yes	string	
`version_id`	path	Yes	string	
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

POST /prompts/{agent_type}/rollback

Rollback Prompt

Roll back the active prompt N steps.

Parameters

Name	In	Required	Type	Description
`agent_type`	path	Yes	string	
`steps`	query	No	integer	
`X-API-Key`	header	No	string	

Responses

Code	Description
200	Successful Response
422	Validation Error

Schemas

ApprovalRequestResponse

Property	Type	Description

`request_id`	string	
`run_id`	string	
`tenant_id`	string	
`tool_name`	string	
`tool_args`	object	
`reason`	string	
`status`	string	
`created_at`	string	
`expires_at`	string	
`resolved_at`	union	
`resolved_by`	union	

CreateRunRequest

Request body for creating a new agent run.

Property	Type	Description
`agent_type`	string	Runnable native agent type: sql or code
`task`	string	The task for the agent to execute
`metadata`	object	Optional metadata

ErrorRecordResponse

Property	Type	Description
`record_id`	string	
`agent_type`	string	
`task`	string	
`failure_class`	string	
`error_message`	string	
`stack_trace`	string	
`created_at`	string	

EvalCompareRequest

Optional controls for a prompt comparison.

Property	Type	Description
`baseline_prompt_version`	string	
`patched_prompt_version`	string	

EvalRunRequest

Optional controls for an eval run.

Property	Type	Description
`prompt_version`	string	Prompt version label

GraphQueryResponse

Response for graph traversal queries.

Property	Type	Description
`paths`	array	
`context`	string	

HTTPValidationError

Property	Type	Description
`detail`	array	

MemoryEntryResponse

A single retrieved memory entry.

Property	Type	Description
`id`	string	
`text`	string	
`metadata`	object	
`score`	number	
`source`	string	

PatchOutcomeResponse

Property	Type	Description
`patch_id`	string	
`baseline_score`	number	

patched_score	number	
improvement	number	
accepted	boolean	
eval_summary	string	

PatchResponse

Property	Type	Description
patch_id	string	
agent_type	string	
target	string	
op	string	
path	string	
value	string	
rationale	string	
proposed_by	string	
proposed_at	string	
score	union	
status	string	
based_on_errors	array	

PromptVersionResponse

Property	Type	Description
version_id	string	
agent_type	string	
content	string	
version_number	integer	
active	boolean	
score	union	
patch_id	union	
created_by	string	
created_at	string	

`tags`	array	
`metadata`	object	

RecallRequest

Request body for retrieving memory entries.

Property	Type	Description
`query`	string	Search query
`k`	integer	Number of results
`filter`	object	Metadata filter

RememberRequest

Request body for storing a memory entry.

Property	Type	Description
`text`	string	Text to store
`metadata`	object	Optional metadata

RememberResponse

Response for a remember operation.

Property	Type	Description
`id`	string	

ResolveRequest

Property	Type	Description
`resolved_by`	string	Username or system ID of resolver

RunRecordResponse

API response model for a RunRecord.

Property	Type	Description
`run_id`	string	
`tenant_id`	string	
`agent_type`	string	
`task`	string	

`status`	string	
`result`	union	
`created_at`	string	
`started_at`	union	
`completed_at`	union	
`hitl_pending`	boolean	
`metadata`	object	

ValidationError

Property	Type	Description
`loc`	array	
`msg`	string	
`type`	string	
`input`	string	
`ctx`	object	